

1.1)

a) $\|\vec{u} + \vec{v}\|^2 = (\sqrt{\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle})^2$ Definice normy

$$= \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle$$

$$= \langle \vec{u} + \vec{v}, \vec{u} \rangle + \langle \vec{u} + \vec{v}, \vec{v} \rangle \quad \text{- linearity v druhé složce}$$

$$= \langle \vec{u}, \vec{u} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{v} \rangle \quad \text{- linearity v první složce}$$

$$= \|\vec{u}\|^2 + \overline{\langle \vec{u}, \vec{v} \rangle} + \langle \vec{u}, \vec{v} \rangle + \|\vec{v}\|^2 \quad \text{- skokosymetrie}$$

$$= \|\vec{u}\|^2 + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2 \quad \text{- } z + \bar{z} = 2\operatorname{Re}(z)$$

$\|\vec{u} - \vec{v}\|^2 = (\sqrt{\langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle})^2$ Definice normy

$$= \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle$$

$$= \langle \vec{u} - \vec{v}, \vec{u} \rangle - \langle \vec{u} - \vec{v}, \vec{v} \rangle \quad \text{- linearity v druhé složce}$$

$$= \langle \vec{u}, \vec{u} \rangle - \langle \vec{v}, \vec{u} \rangle - (\langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{v} \rangle) \quad \text{- linearity v první složce}$$

$$= \|\vec{u}\|^2 - (\overline{\langle \vec{u}, \vec{v} \rangle} + \langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2 \quad \text{- skokosymetrie}$$

$$= \|\vec{u}\|^2 - 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2 \quad \text{- } z + \bar{z} = 2\operatorname{Re}(z)$$

$$\frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2) = \frac{1}{4} (\|\vec{u}\|^2 + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2 - (\|\vec{u}\|^2 - 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2)) = \frac{1}{4} \cdot 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

b) $\|\vec{u} + i\vec{v}\|^2 = (\sqrt{\langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle})^2$ Definice normy

$$= \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle$$

$$= \langle \vec{u} + i\vec{v}, \vec{u} \rangle + \langle \vec{u} + i\vec{v}, i\vec{v} \rangle \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{linearity v druhé složce}$$

$$= \langle \vec{u}, \vec{u} \rangle + \langle i\vec{v}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle + i\langle i\vec{v}, \vec{v} \rangle \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{linearity v první složce}$$

$$= \langle \vec{u}, \vec{u} \rangle - i\langle \vec{v}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle - i^2\langle \vec{v}, \vec{v} \rangle$$

$$= \|\vec{u}\|^2 - i\overline{\langle \vec{u}, \vec{v} \rangle} + i\langle \vec{u}, \vec{v} \rangle + \|\vec{v}\|^2 \quad \text{- skokosymetrie}$$

$$= \|\vec{u}\|^2 + i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) + \|\vec{v}\|^2$$

Označme $\langle \vec{u}, \vec{v} \rangle = a + bi$. Potom:

$$\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} = a + bi - (a - bi) = 2bi = 2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

$$= \|\vec{u}\|^2 + 2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2$$

$\|\vec{u} - i\vec{v}\|^2 = (\sqrt{\langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle})^2$ Definice normy

$$= \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle$$

$$= \langle \vec{u} - i\vec{v}, \vec{u} \rangle - \langle \vec{u} - i\vec{v}, i\vec{v} \rangle \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{linearity v druhé složce}$$

$$= \langle \vec{u}, \vec{u} \rangle - \langle i\vec{v}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle - i\langle i\vec{v}, \vec{v} \rangle \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{linearity v první složce}$$

$$= \langle \vec{u}, \vec{u} \rangle + i\langle \vec{v}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle + i^2\langle \vec{v}, \vec{v} \rangle$$

$$= \|\vec{u}\|^2 + i\overline{\langle \vec{u}, \vec{v} \rangle} - i\langle \vec{u}, \vec{v} \rangle - \|\vec{v}\|^2 \quad \text{- skokosymetrie}$$

$$= \|\vec{u}\|^2 - i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) + \|\vec{v}\|^2$$

Označme $\langle \vec{u}, \vec{v} \rangle = a + bi$. Potom:

$$\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} = a + bi - (a - bi) = 2bi = 2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

$$= \|\vec{u}\|^2 - 2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2$$

$$\frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) = \frac{1}{4} (\|\vec{u}\|^2 + 2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2 - (\|\vec{u}\|^2 - 2i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) + \|\vec{v}\|^2)) = \frac{1}{4} \cdot 4i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = i\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$